

3. Implement Future Sales Prediction using a suitable machine learning algorithm.

CODE:

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[14] # Future Sales Prediction using Linear Regression
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Sample Advertising Dataset
data = {
    'TV': [230.1,44.5,17.2,151.5,180.8,8.7,57.5,120.2,199.8,66.1],
    'Radio': [37.8,39.3,45.9,41.3,10.8,48.9,32.8,19.6,2.6,5.8],
    'Newspaper': [69.2,45.1,69.3,58.5,58.4,75.0,23.5,11.6,21.2,24.2],
    'Sales': [22.1,10.4,9.3,18.5,12.9,7.2,11.8,13.2,10.6,8.6]
}
df = pd.DataFrame(data)
print("Dataset")
print(df)
# Features and Target
X = df[['TV', 'Radio', 'Newspaper']]
y = df['Sales']
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
# Create Model
model = LinearRegression()
# Train Model
model.fit(X_train, y_train)
# Predict Test Data
y_pred = model.predict(X_test)
# Evaluation
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

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[14] print("\nModel Performance")
print("-----")
print("Mean Squared Error :", mse)
print("Root Mean Squared Error :", rmse)
print("R2 Score :", r2)
# Actual vs Predicted
result = pd.DataFrame({
    'Actual Sales': y_test.values,
    'Predicted Sales': y_pred
})
print("\nActual vs Predicted")
print(result)
# Future Prediction
future_data = pd.DataFrame({
    'TV':[200],
    'Radio':[25],
    'Newspaper':[40]
})
future_sales = model.predict(future_data)
print("\nPredicted Future Sales =", future_sales[0])
# Graph
plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred, color='blue', s=80)
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    color='red',
    linewidth=2
)
plt.title("Actual vs Predicted Sales")
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.grid(True)
plt.show()
```

OUTPUT:

```
Dataset
***  TV   Radio  Newspaper  Sales
0    230.1  37.8    69.2    22.1
1    44.5   39.3    45.1    10.4
2    17.2   45.9    69.3     9.3
3    151.5  41.3    58.5    18.5
4    180.8  10.8    58.4    12.9
5     8.7   48.9    75.0     7.2
6    57.5   32.8    23.5    11.8
7    120.2  19.6    11.6    13.2
8    199.8   2.6    21.2    10.6
9     66.1   5.8    24.2     8.6

Model Performance
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Mean Squared Error : 14.424764626878826
Root Mean Squared Error : 3.797994816594518
R2 Score : -1441.4764626878928

Actual vs Predicted
  Actual Sales  Predicted Sales
0             10.6      15.956130
1             10.4      10.801742

Predicted Future Sales = 19.028323869951908
```

